Lesson plan · for the teacher

Cleaning and organising data: unit check

Data engineering · Level 2 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Tidy sheets a computer can use
- Names that agree
- Blanks, zeros and repeats
- One ID per pupil: joining sheets

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which sheet in our school would you clean first?
- What would a master list change for us?
- Which check would have caught our worst mistake?

Exit ticket

- One cleaning habit I will use this week, and on which sheet.

Try it in class tomorrow
Take one real sheet your school uses, make a tidy copy with pupil IDs, run the four checks on it and write a short cleaning log.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Where do formulas such as AVERAGE and COUNTIF belong?**

- ◯ On a separate summary sheet that reads the data
- ◯ In a Total row under the pupils
- ◯ Typed over the raw marks

2 **Python reads every value in a CSV file as text, so marks need int() before they can be added.**

- ◯ True
- ◯ False

3 **Which of these would a computer count as different from the name "Grace"? Choose all that apply.**

- ☐ "grace"
- ☐ " Grace" (with a space at the start)
- ☐ "Grace"
- ☐ "Grayce"

4 **Put the cleaning steps in order.**

- ☐ Log what you changed
- ☐ Remove true repeats
- ☐ Match names to the master list
- ☐ Tidy spaces and capitals
- ☐ Keep the raw sheet untouched

5 **Four maths marks are 12, 16, blank and 8. What should the report say?**

- ◯ The average is 9, counting the blank as 0
- ◯ The average of 3 marks is 12; 1 mark is missing
- ◯ The average is 12, and say nothing about the blank